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Rational Design of 2D h-BAs Monolayer as Advanced Sulfur hosts for High Energy Density Li-S Batteries

Nabil Khossossi^{1,2}, Pritam Kumar Panda¹, Deobrat Singh¹, Vivekanand Shukla³, Yogendra Kumar Mishra⁴, Ismail Essaoudi², Abdelmajid Ainane^{1,2,6}, Rajeev Ahuja^{*1,5}

¹Condensed Matter Theory Group, Department of Physics and Astronomy, Uppsala University, 75120 Uppsala, Sweden

²Laboratoire de Physique des Matériaux et Modélisations des Systèmes, (LP2MS), Unité Associée au CNRST-URAC 08, Moulay Ismail University, Faculty of Sciences, Department of Physics, B.P. 11201, Meknes, Morocco

³Department of Microtechnology and Nanoscience (MC2), Chalmers University of Technology, SE-412 96 Gothenburg, Sweden

⁴NanoSYD, Mads Clausen Institute, University of Southern Denmark, Alsion 2, 6400, Sønderborg, Denmark

⁵Applied Materials Physics, Department of Materials and Engineering, Royal Institute of Technology (KTH), S-100 44 Stockholm, Sweden

⁶Max-Planck-Institut für Physik Complexer Systeme, Nöthnitzer Str. 38, D-01187 Dresden, Germany

*Corresponding author: rajeev.ahuja@physics.uu.se

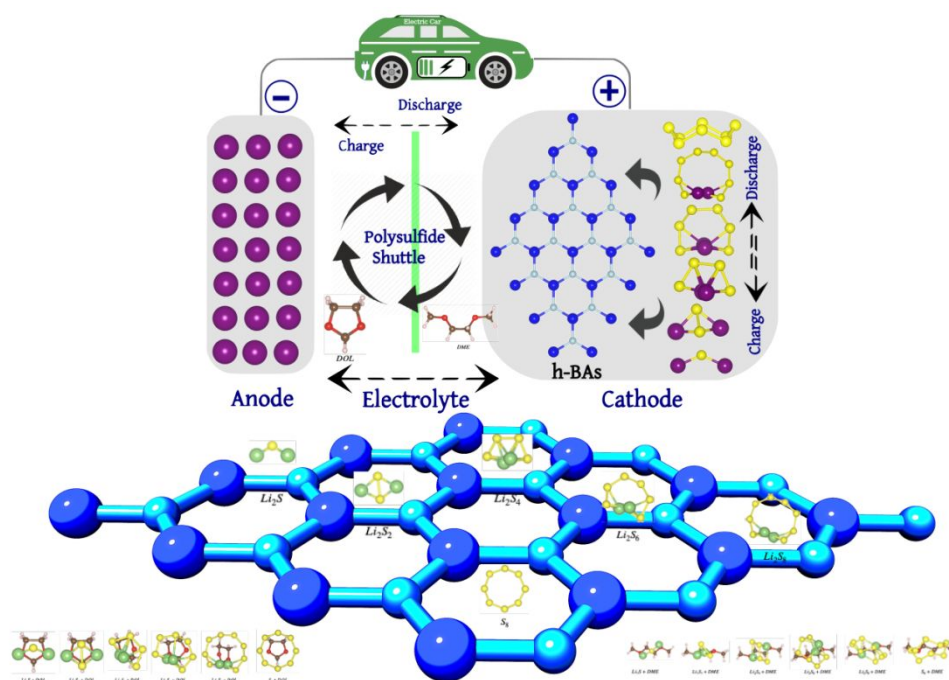
Abstract

The emergence of compact Lithium-Sulfur (Li-S) batteries with improved performances is becoming one of the most desirable aspects of future energy technologies. Beyond Li-ion batteries, Li-S is of great relevance to follow that adapts to the specificity of each application. It is among the most suitable elements for high-performance energy storage systems given its high theoretical capacity (1674 mAhg⁻¹) and the energy density (2600 Whkg⁻¹) relative to Li-ion batteries (300 Whkg⁻¹). Nevertheless, the high-cell polarization and the shuttle effect constitute an enormous challenge towards the concrete applications of Li-S batteries. In the framework of this work, the functional theory of density (DFT) calculations have been carried out to analyze the potential of h-BAs nanosheet as a promising host material for Li-S batteries. Binding and electronic characteristics of lithium polysulfides (LiPSs) adsorbed on h-BAs surface have been explored. Reported findings highlight the potential of h-BAs monolayer as a moderate host material given the binding energies of different LiPSs varies from 0.47 eV to 3.55 eV. A more detailed analysis of the complex binding mechanisms is carried out by investigating the components of van-der-Waals (vdW), physical/chemical interactions. The

defected surface of h-BAs monolayer has optimum binding energies with LiPSs for Li-S batteries. All these findings provide valuable insights into the binding and electronic characteristics of h-BAs monolayer as a moderate host material for Li-S batteries.

Keywords: Lithium-polysulfides; DFT; Boron Arsenide; Adsorption; Shuttle effect; Nudge-elastic band.

Graphical Abstract



1. Introduction

Human dependence on energy - consuming resources involves improving technologies to solve the global energy crisis ¹. In an effort to mitigate the challenges of curbing CO₂ emissions, harnessing renewable energy sources marked a revolution to switch towards electric vehicles, smart grids, portable electronic devices ². Such a drastic change in utilizing the power of electrical energy, e.g., lithium-ion batteries, that power our lives and can be accessed with ease and efficiency in a potentially healthier environment ³. In further instances, the advances on Li-ion batteries have revolutionized the science of energy storage and rendered the mobile revolution feasible by its high potential, high energy density and efficiency ^{4,5}. Despite several developments, the search for efficient electrode materials is very daunting and challenging in general ⁶. Lithium-sulfur batteries that are low-cost, high energy density and lower toxic than current stocks will constitute the alternate or viable option for existing products^{7,8}. Exploring battery chemistry, e.g., Li-S, typical strategies include optimization of electrode materials (in our case, 2D Boron Arsenide monolayer) with higher capacity and potential. Current battery technology yields less than 400 W/kg of gravimetric energy density, which is based on the insertion reaction mechanism ^{9,10}. In order to achieve more capacity and potential, sulfur redox batteries based on multiple electron reactions seem interesting, provided that sulfur is very cheap and strong, making battery production easier ¹⁰. Moreover, the two-electron redox site provided by sulfur in conjunction with Li metal gives a tremendous theoretical specific capacity of 1672 mAh/g and energy density of 2600 Wh/kg ^{8,11}.

Li-S batteries undergo complex conversion reactions between metallic Li and elemental sulfur as compared to the reversible incorporation and extraction of Li⁺-ions in typical LIBs. During the discharge process, elemental sulfur is electrochemically reduced at the cathode to polysulfides ¹². Similarly, during the charging process, Li₂S converted into soluble short-chain (Lithium polysulfides) LiPSs and then into long-chain LiPSs and further to solid S₈ phase ¹³. The electrochemical process that makes Li-S batteries more effective is highly reliable on the performance of electrode materials. Since 2D materials have been proposed to be effective in many physical and chemical applications, it will be worth-mentioning the role of 2D materials as it possesses unique features in guiding next-generation Li-S batteries ¹⁴.

Despite these major advantages, a number of problems, including sulfur isolating, a large - volume change and the main issue – the dissolution and the shuttling effect of intermediate soluble LiPSs still greatly hinder the commercialized process of this promising material ⁵⁷. The manufacturing of various sulfur–carbon composites or rational sulfur electrode systems focuses

on different methods for the purpose to reduce the diffusion of LiPSs in liquid electrolytes from cathodes⁵⁸. Furthermore, by means of physical adsorption techniques, the most conventional method for the reduction of polysulfides is to include sulfur in the hollow/porous conducting host. Because of the weak van der Waals interaction, the physical absorption mechanism cannot sustain for long term as it cannot restrict the diffusion of lithium polysulfides. The intentional loss of active materials and "shuttle effect" thus could only be minimized under this physical protection, but not eradicated⁵⁷.

During the past decade, several thousands of 2D materials have been experimentally synthesized or theoretically predicted for various applications^{15,16}. To name a few, phosphorene, graphenylene, borophene, BC₂N, Be₃N₂, Mo₂C, Ti₃C₂, GeS, Sc₂C, VS₂, SiC, and VC₂ can be regarded as efficient materials by their virtue of excellent energy storage capacities^{17–21}. The intuitive picture of enhancing the conversion kinetics and promotion effect can be modulated using bidimensional materials such a sulfur-deficient molybdenum disulfide surface²² that involved lithiation-cleavage mechanism which provides great thermodynamics as well as kinetic driving interaction for Li₂S_x LiPSs conversion. Yu et al.,²³ depicted Boron-Phosphide monolayer as a promising host material because of the adsorption energy for Li₂S_x LiPSs on the BP monolayer are between of 0.57–2.85 eV by analyzing the components of van der Waals, electrostatic and chemical interactions. Recently, Mxene-engineered Li-S batteries have been developed, improving sulfur utilization and cycling stability^{24–29}. Moreover, Transition metal oxides VO₂, sulfides VS₂, and embedded C₂N have demonstrated excellent properties, including moderate adsorption strength for Li₂S_x, faster Li⁺ transport, and accelerated surface redox reaction^{19,30}. Zhao et al.³¹ studied SiC₂ siligraphene as moderate host substrate for Li-S batteries that revealed comparatively not great adsorption energy for soluble Li₂S_x LiPSs due to S-Si and Li-C interactions. The implementation of dopings effect and vacancies can improve the adsorption energies of different Li₂S_x LiPSs with BN monolayers due to the great Li...N or S...B interaction that moderately interacted with BN nanosheets preserving a balance between adsorption strength and integrity³².

Despite several achievements, the quest for 2D materials for energy storage applications is still demanding, especially for rechargeable batteries³³. A good deal of theoretical research effort based on DFT, and experiment studies, was reported on single-layer graphene-like materials made of binary elements of groups III-V³⁴. In comparison to h-BP, which already investigated as a moderate host material for Li-S batteries²³, h-BAs monolayer has a smaller bandgap and

served as a promising alternative for anode material in the case of alkali based batteries^{35,36}. Nevertheless, to the best of our knowledge, the interaction of Lithium and Sulfur with h-BAs has not been articulated. And in order to fill the discrepancies. By adopting the first-principles-approach, we have explored h-BAs atomic thick layer for Li-S battery application. Initially, we began by evaluating the structural and electronic properties of h-BAs monolayer and S-containing species. After that, the binding strength of S₈/Li₂S_x LiPSs at different possible configuration sites and orientation on h-BAs monolayer as well as on the organic liquid electrolytes DOL/DME were calculated. The distribution of differences of spatial charge densities were illustrated. We investigated the binding strength from chemisorption and vdW-physisorption to better understanding the behavior of the underlying anchoring mechanisms and the diffusion and catalytic decomposition of Li₂S over the h-BAs surface were evaluated. Finally, for the purpose of improving the adsorption of Li₂S_x (x=8,6,4) LiPSs, the interaction between Li₂S_x and defected systems of the h-BAs host material was also investigated and compared with the pristine monolayer.

2. Methodology

The structure optimizations and electronic calculations are carried out based on the generalized gradient approximation developed by Perdew et al. as implemented in the Quantum_espresso package.^{37,38,39,40} The projected augmented waves pseudo-potentials were used with a plane-wave cut-off of 500 Ry. The lattice parameter and atomic positions of pristine h-BAs monolayer before and after the polysulfides adsorption are relaxed with energy and forces of 10⁻⁶ eV and 10⁻² eV/Å, respectively. A Monkhorst pack grid⁴¹ of 5×5×1 K-mesh were considered for a 4×4×1 supercell h-BAs pristine system and after the S₈/Li₂S_x polysulfides adsorption.

The binding strength of different S-containing molecules adsorbed on the h-BAs monolayer provide a critical factor in enhancing the capabilities of Li-S battery. The equilibrium adsorption sites were obtained by placing different S-containing molecules on all available sites with different orientations and comparing the resulting adsorption energies of S₈/Li₂S_x molecules, E_{ads} defined as:

$$E_{\text{ads}} = E_{\text{Substrate}} + E_{\text{S}_8/\text{Li}_2\text{S}_x} - E_{\text{System}} \quad (1)$$

Here, $E_{\text{Substrate}}$ represent the energy of pristine system, $E_{\text{S}_8/\text{Li}_2\text{S}_x}$ represents the energy of isolated S₈/Li₂S_x molecules in the bulk reference state with a large cubic lattice parameter (a=b=c=20Å), and E_{System} refer to the energy of the absorbed state. By definition, negative

adsorption energy refers to a repulsion of the interacting polysulfides and host material. On the other hand, a positive value E_{ads} signify attraction, indicating that the adsorption site is suitable and stable. The diffusion barrier and catalytic decomposition of different $\text{S}_8/\text{Li}_2\text{S}_x$ polysulfides on h-BAs nanosheet were investigated with the CI-NEB method^{42,43} by using two most nearest stable adsorption configurations as initial and final positions with 7 intermediates states generated by linear interpolation. The effect of van der Waals interaction was described by using the empirical correction in the Grimme method (DFT-D3) on both geometry optimization and $\text{S}_8/\text{Li}_2\text{S}_x$ polysulfides adsorptions.^{45,46}

To assess the stability of the h-BAs monolayer together with its feasibility for experimental synthesis, the cohesive energy for the pristine system is calculated according to the formula :

$$E_{\text{coh}} = \frac{(E_{\text{BAs}} - n_{\text{As}}E_{\text{As}} - n_{\text{B}}E_{\text{B}})}{(n_{\text{As}} + n_{\text{B}})} \quad (2)$$

where, E_{BAs} , represent the energy of pristine h-BAs monolayer, E_{As} and E_{B} refers to the energies of isolated As-, and B-atoms, respectively. The n_{As} and n_{B} refer to the number of As- and B-atoms in the system, respectively. Furthermore, the formation energies of defect systems compared to a pure h-BAs monolayer are also computed using the following formula:

$$E_{\text{formation}} = E_{\text{defect}} - E_{\text{pure}} + \sum n_i u_i \quad (3)$$

Where, E_{defect} and E_{pure} are the total energies of the h-BAs nanosheet in the presence and absence of vacancies, respectively. The n_i and u_i with $i = \text{As, B}$ represent the chemical potential and the number of removed As and B atom from h-BAs nanosheet, respectively. To re-check the thermal stability of the pristine monolayer before and after the adsorption of $\text{S}_8/\text{Li}_2\text{S}_x$ polysulfides, AIMD simulations were conducted as part of the NVT ensemble using VASP package^{47,48} by keeping the temperature fixed at 300K for a timescale of 10ps and MD-step of 2fs. Additional detailed information of the computational methods can be found on our most recent works.^{44,59,60}

3. Results and discussion

3.1. Structural properties of h-BAs monolayer and S-containing species.

Initially, we have focused on the structural characteristics of the pristine h-BAs monolayer and S-containing species. Figure 1a illustrates the top view of the freestanding ($4 \times 4 \times 1$) supercell

of h-BAs monolayer after full relaxation, and the dotted line shows the unit cell contains two atoms. It is evident that h-BAs monolayer exhibits a hexagonal planar surface corresponding to the space group P-6m2 (187) similar to that of 2D hexagonal graphene. In h-BAs monolayer, Both B- and As-atoms are located at the same height in z-directions, wherein each B- or As-atom contains three coordination numbers. The fully relaxation of the BAs system provides a structural parameters, $a = b = 3.392 \text{ \AA}$ with a vacuum layer fixed to $c = 20 \text{ \AA}$. The $d_{\text{B-As}}$ about $1.958 \text{ \AA}$ and angle length is 120° , all these findings are in line with previous reports ^{34,35,49}.

To evaluate the energetic, dynamic, and the high temperature stability of the pristine h-BAs before the adsorption of $\text{S}_8/\text{Li}_2\text{S}_x$ polysulfides. Based on our calculations, the cohesive energy of pristine h-BAs monolayer calculated by the equation (2) is about -5.42 eV per atom, implying the energetic stability of h-BAs nanosheet. Furthermore, Phonon spectrum computation was carried out within the framework of the density functional perturbation theory implemented in Phonopy package ⁵⁰, the figure S1 of the SI shows the dynamic stability of the system with the absence of any negative frequencies. Finally, The AIMD simulations of the pristine system is carried out, Figure S3 and the animation video A1 in SI demonstrate the high temperature stability of the structure.

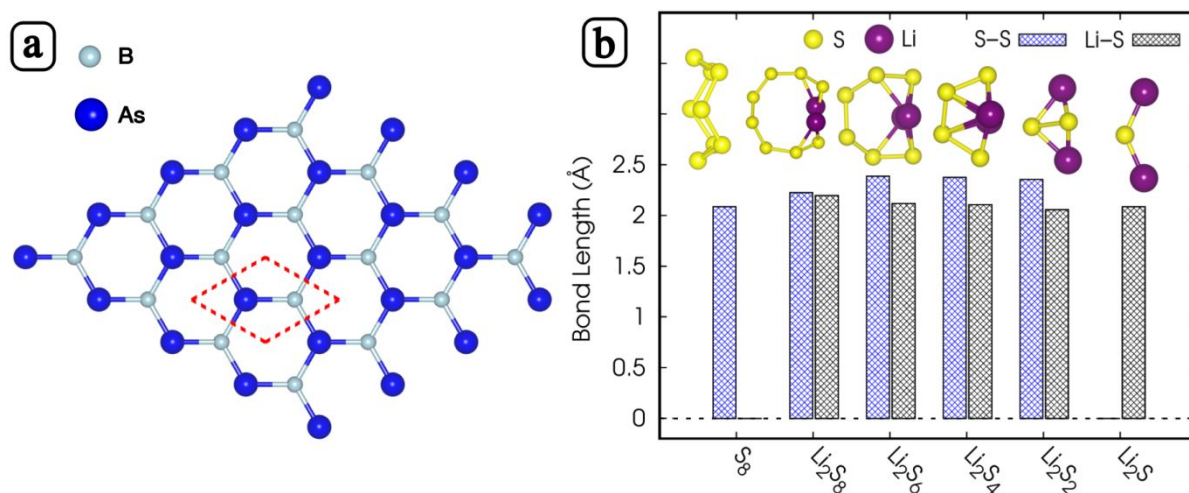


Figure 1. (a) The optimized structures of the h-BAs nanosheet. (b) The bond length Li-S and S-S and the screenshots of relaxed molecular structures S_8 and Li_2S_x ($x=1,2,4,6,8$).

In contrast to the commercialized Li-Ion batteries, A Li-S Batteries are essentially consist of a metallic-Li anode (Instead of the intercalation anode in LIBs) and a sulfur cathode separated by an electrolyte. Through the discharge-process, the metallic Li-anode is oxidized, producing the Li^+ ions, which like in LIB, let behind the electrons at the negative electrode. Thus, the electrons shift over an external circuit towards the cathode to power the electronic device, while

lithium-ion moves across porous separator heads for the cathode in which it accumulates together with different sulfur concentrations to progressively produce lithium polysulfide Li_2S_x ($x=1,2,4,6,8$) molecules. These polysulfides are crucial phases of the lithiation process, as documented through both practical and computational research⁵¹.

Before moving to a detailed investigation of h-BAs as a moderate host material for Li-S batteries, it is thus important to optimize different S-containing molecules, which include S_8 and Li_2S_8 , Li_2S_6 , Li_2S_4 , Li_2S_2 , Li_2S polysulfides. All these S-containing species exhibit 3D configurations instead of 1D chain configurations (with lithium bonded at the horns)^{51,52}. Figure 1b and Figure S4 in SI depicts the bond length of Li-S and S-S as well as the relaxed molecular structures of S_8 and Li_2S_x ($x=1,2,4,6,8$). It is clearly evident that the Li-S and S-S bond lengths vary with the decreasing Sulfur concentration in Li_2S_x ($x=1,2,4,6,8$) molecules, suggesting that the long-chain Li_2S_x ($x=4,6,8$) polysulfides are easier to ionize into Li^+ and LiPSs anions in the electrolyte than the Li_2S_2 or Li_2S ³².

3.2. Adsorption of polysulfides on the h-BAs surface

Proceeding with our analysis, we focus on the interaction between $\text{S}_8/\text{Li}_2\text{S}_x$ molecules and h-BAs surface. Against this background, potentially efficient host materials for Li-S batteries is expected to be distinguished by a relatively high binding energy, which generally contributes to the loading of S_8 and Li_2S_x molecules on the surface. One can calculate these binding energies with/without van-der-Waals interactions. For a deeper understanding of suitable adsorption sites on the h-BAs monolayer, we start assessing the binding energies by considering different initial orientations of S-containing species at different starting sites on the h-BAs surface in a relatively large (4×4)-supercell. Figure 2(a-f) demonstrates the top and side views of fully relaxed $\text{S}_8/\text{Li}_2\text{S}_x$ molecules adsorbed at the most stable sites and orientations on the h-BAs surface.

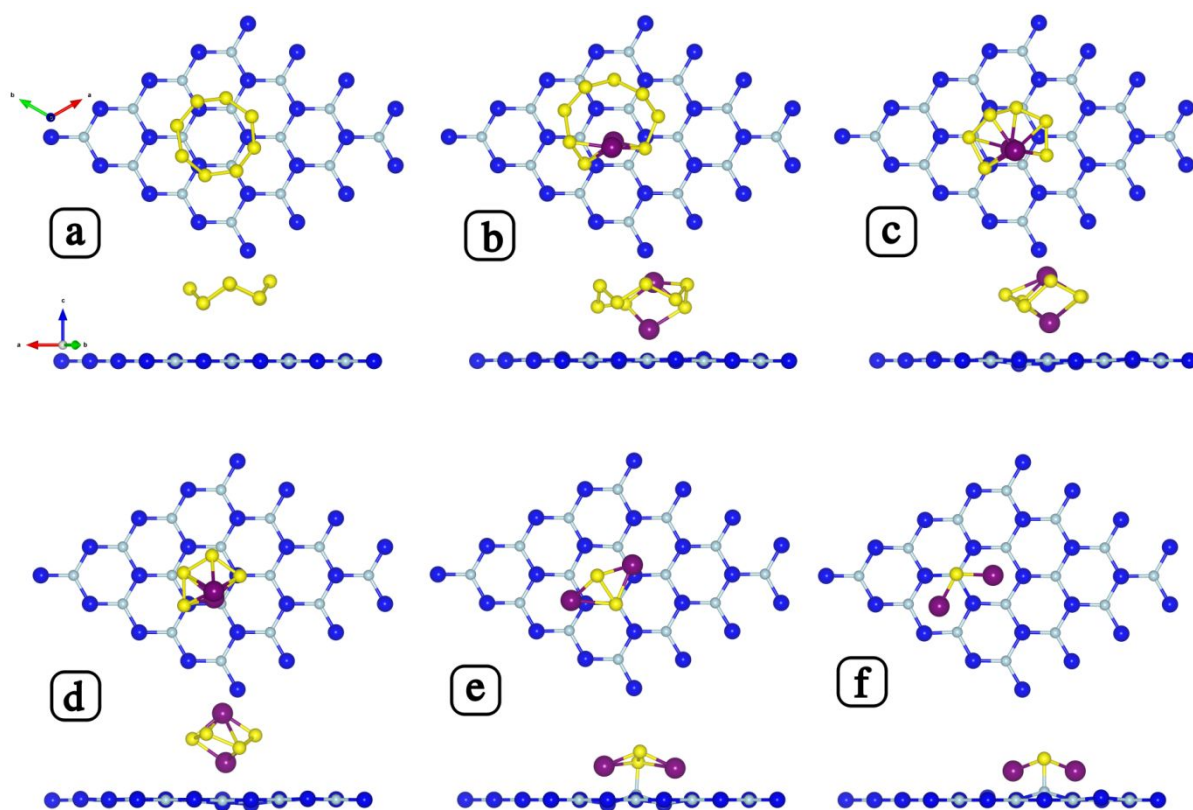


Figure. 2. Top and side views strongest configurations of LiPSs (a) S_8 , (b) Li_2S_8 , (c) Li_2S_6 , (d) Li_2S_4 , (e) Li_2S_2 , (f) Li_2S molecules on the h-BAs surface.

It is clear that the S_8 molecule relaxed parallel to the h-BAs surface after full relaxation within a medium distance of about 4.09 Å between Sulfur atom of S_8 molecule and Arsenic of h-BAs, while the S_8 and h-BAs structures remain the same initial geometry without any modification, suggesting that no obvious chemical bond is formed (Physisorption of the S_8 molecule). In the case of Li_2S_x polysulfides, we have two categories, the first one is Li_2S_x (with $x=4,6,8$), where S-atoms generally appear to be horizontal to h-BAs surface, and both or either of Li-atoms are lying on the hollow of a B_3As_3 -ring over the h-BAs surface. The second one is Li_2S_x (with $x=1,2$) at which the Li-atoms are situated at approximately above As-atoms for Li_2S_2 , and at the hollow site for Li_2S with S-atom above the B-atom. The calculated adsorption distances between Li-atom and h-BAs surface are about 2.04 and 1.84 Å for Li_2S_2 and Li_2S , respectively. Typically, as the concentration of S-atom in Li_2S_x decreases, The adsorption separation between S_8/Li_2S_x and h-BAs surface lie between 4.09 and 1.84 Å can be found in table 1. It is perceptible from table 1 and table S1 in SI, that there is a slight deformation/distortion in the S_8/Li_2S_x molecular structures after their adsorption on the h-BAs monolayer. S-S/Li-S bond-length differences in S_8/Li_2S_x molecules before and after their adsorption on h-BAs are

computed and also summarized in table 1. It must be noted that the S-S bond-lengths are decreased, and Li-S bond-lengths are increased in all the polysulfides during the adsorption process. Additionally, the wider bond-lengths modification occurs in Li_2S_2 and Li_2S compared to the rest Li_2S_x (with $x=4,6,8$) molecules. This may transpire due to the lower number of lithium and sulfur bonds in Li_2S_2 and Li_2S , which makes it easier to expand the $d_{\text{Li-S}}$ bond-length.

Table. 1. The adsorption energies (E_{ads}) of $\text{S}_8/\text{Li}_2\text{S}_x$ adsorbed on the h-BAs nanosheet, binding energies (E_b) of the same with electrolyte molecules (DOL and DME liquid organic electrolytes), respectively. The estimated average distances were separating $\text{S}_8/\text{Li}_2\text{S}_x$ and h-BAs surface ($d_{\text{Clusters-BAs}}$) and the average difference of Li-S ($\Delta d_{\text{Li-S}}$) and S-S ($\Delta d_{\text{S-S}}$) bond-lengths. At last ΔQ_{BAs} , ΔQ_{Li} and ΔQ_{S} is the charge transfer of pristine h-BAs monolayer, Li- and S-atoms of $\text{S}_8/\text{Li}_2\text{S}_x$.

	S_8	Li_2S_8	Li_2S_6	Li_2S_4	Li_2S_2	Li_2S
E_{ads}	0.46	1.07	0.83	0.77	2.71	3.52
$E_{b(\text{LiPSs} + \text{DOL})}$	0.18	0.78	0.81	0.79	1.07	0.90
$E_{b(\text{LiPSs} + \text{DME})}$	0.20	0.95	1.16	0.96	1.11	0.98
$d_{\text{Clusters-BAs}}$	4.09	3.50	3.78	3.65	2.04	1.84
$\Delta d_{\text{Li-S}}$	--	0.012	0.046	0.079	0.126	0.205
$\Delta d_{\text{S-S}}$	0.009	-0.008	0.004	-0.039	-0.087	--
ΔQ_{BAs}	-0.02	0.09	0.21	0.38	0.52	0.67
ΔQ_{Li}	--	0.00	0.00	-0.02	-0.02	-0.04
ΔQ_{S}	-0.02	0.09	0.21	0.36	0.50	0.63

The most critical parameter, which governs the anchoring capabilities of any host material, is the binding energy. These binding energies (E_{vdW}) contributed by physical-interaction of various $\text{S}_8/\text{Li}_2\text{S}_x$ molecules over the h-BAs surface are computed by using the equation⁵³:

$$E_{vdW} = E_{with - vdW} - E_{without - vdW} \quad (4)$$

Where, $E_{with - vdW}$ and $E_{without - vdW}$ represent the binding energy computed in the presence and absence of the vdW correction contribution, respectively. It is obvious from figure 3a, that the physical components of the binding energies are positive, which demonstrate the attractive characteristic of h-BAs monolayer towards all S_8/Li_2S_x molecules. The binding energies of S_8 and Li_2S_x ($x=8,6,4,2,1$) are computed as 0.46 eV, 1.07 eV, 0.83 eV, 0.77 eV, 2.71 eV and 3.52 eV respectively. For the purpose of comparison, these values are significantly greater than those obtained in the case of pristine graphene, C_3B , and C_3N monolayers^{53,54}.

Furthermore, we have systematically investigated the anchoring efficiency of h-BAs through a comparison of the binding energies (E_b) between S_8/Li_2S_x and the electrolyte molecules 1,3-dioxolane (DOL)/dimethyl ether (DME) (1: 1, v/v) mixture, which is usually considered as an accommodating choice for sulfur cathodes²⁷. Both electrolytes commonly used as the liquid organic electrolytes of Li-S batteries in which DOL and DME are present in equal compositions (1:1 mixture of DOL and DME) (See figures S5 and S6 in SI). The binding energies E_b of the S_8/Li_2S_x with the selective electrolytes were evaluated as follows:

$$E_b = E_{total} - E_{DOL/DME} - E_{LiPSs} \quad (5)$$

Where, E_{total} , $E_{DME/DOL}$, and E_{LiPSs} are the computed ground state energies of the bound system of electrolytes and LiPSs, the full optimized LiPSs, and electrolyte molecule, respectively. As illustrated in table 1, the S_8/Li_2S_x molecules on the h-BAs surface are bound with E_{ads} varying from 0.46 to 3.52 eV. On the other hand, it is represented that DOL and DME present binding (E_b) values for the S_8/Li_2S_x molecules from 0.18 to 1.07 eV and 0.20 to 1.16 eV, respectively. These results that the binding energies E_b of the S_8/Li_2S_x molecules with electrolytes are lower compared to those on h-BAs for every reduction step of the LiPSs except Li_2S_4 and Li_2S_6 . From our findings, it can be concluded that the S_8/Li_2S_x molecules would be anchored on h-BAs monolayer, and they would not be dissolved into the electrolyte from the energetics point of view except for the case of Li_2S_4 and Li_2S_6 . Consequently, the considered h-BAs monolayer does not qualify to suppress the shuttle effect. It is hard to get the pure 2D sheet in the experiment. Hence, to get the more realistic picture, we have also investigated the defective h-BAs sheet, which enhanced the binding strength of LiPSs with h-BAs monolayer, and are larger than those of DOL/DME electrolytes with LiPSs. Therefore, the defective h-BAs monolayer

can effectively suppress Li_2S_x shuttling and can prove to be an ideal and moderate host material for Li-S batteries.

Through the lithiation-process, the binding energy in the presence of vdW-interaction is the most prevalent. For instance, the S_8 and Li_2S_x with $x = 4, 6, 8$ favor the formation of overlying configurations above the h-BAs surface, which significantly enhances the physical component of the binding energies. In contrast, Li_2S_2 and Li_2S have a characteristic of forming a chemical-bond between S-atom and As-atom in h-BAs monolayer, as shown in figure 2(e-f). This is a clear indication that the chemical composition of the binding energy is more widespread for both molecules, which dominates over the physisorption. Owing to these findings of the binding energy of S_8 and Li_2S_x molecules, it appears that h-BAs monolayer can be a suitable candidate for Li-S battery.

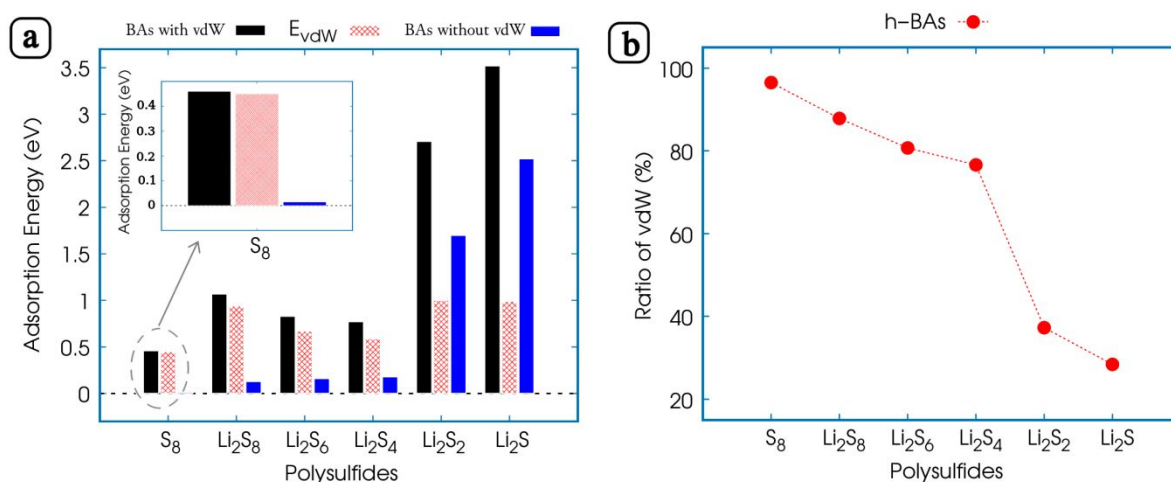


Figure. 3. (a) The computed binding energies with/without vdW-interaction, with the corresponding strength of van-der-Waals physical-interaction for S_8 and Li_2S_x on the h-BAs monolayer. (b) The Ratio of vdW-interaction in binding energies of S_8 and Li_2S_x on the h-BAs monolayer.

The thermal stability of S_8 and Li_2S_x ($x=8, 6, 4, 2, 1$) over pristine h-BAs monolayer between 0 and 10ps at 300 K was performed using MD-simulations. In order to confirm the thermodynamic stabilities, the variation of the total energies versus MD time steps was shown (Figure S3 in ESI), which assures that the total energy is converged in the given time frame for 300 K with very slight variation. Additionally, the molecular dynamic evolution of different species adsorbed on h-BAs monolayer is featured in ESI (Movie B1-6).

To provide a detailed understanding of the underlying anchoring mechanisms as well as to

evaluate the comparative strength of contributions from chemisorption and vdW-physisorption, the ratio of vdW-interaction has been evaluated for different S-containing molecules by :

$$R = \frac{(E_{\text{with} - \text{vdW}} - E_{\text{without} - \text{vdW}})}{E_{\text{with} - \text{vdW}}} \times 100\% \quad (6)$$

Where, $E_{\text{with} - \text{vdW}}$ and $E_{\text{without} - \text{vdW}}$ stand for the binding energy calculated with and without the vdW-interactions with the system set at the DFT-D3 relaxed system, respectively. As already revealed in Figure 3b, the vdW interaction appears superior in the non-lithiated stage with a ratio R about 96.52%, indicating that h-BAs monolayer is unable to create a chemical bond with S8 molecule and the S8-adsorption is governed mainly by the vdW-physical component of binding with the sulfur atom. During the lithiation-process, the physical interaction presents an essential contribution in the binding energy of Li_2S_x ($x=8,6,4$), with a ratio varying between 87.85% and 76.62%. Over time, while the lithiation process proceeds, the physical interaction decreases significantly (Li_2S_2 and Li_2S), and the ratio R drops between 37.27 % and 28.41 %. The variations in adsorption energies of $\text{S}_8/\text{Li}_2\text{S}_x$ molecules (E_{ads}) indicate a significant role of Li-atoms in the chemical-interaction, and S-atoms are involved in the physisorbed interaction. The lowest adsorption energies occurred for S_8 and Li_2S_4 molecules. On the other hand, the h-BAs monolayer displays reasonable adsorption energies towards the Li_2S_x molecules.

To improve our understanding of the difference between the chemical and physical interactions, as well as to elucidate better the reaction of different S-containing molecules and h-BAs monolayer, we have also plotted the difference of spatial-charge-distribution of the different adsorbed polysulfide and the h-BAs monolayer. This can be obtained by employing the formula:

$$\Delta\rho(x,y,z) = \rho_{\text{System}} - \rho_{\text{Substrate}} - \rho_{\text{S8/Li2Sx}} \quad (7)$$

Wherein ρ_{System} , $\rho_{\text{Substrate}}$ and $\rho_{\text{S8/Li2Sx}}$ denote the charge densities for $\text{S}_8/\text{Li}_2\text{S}_x$ on h-BAs surface, pristine BAs monolayer, and isolated $\text{S}_8/\text{Li}_2\text{S}_x$ molecules, respectively. As shown in figure 4(a-f), in the case of non-lithiated S_8 molecule adsorption over the h-BAs surface, it can be clearly identified that there is no charge transfer, resulting in that the adsorption of S_8 molecule is governed practically by the lower vdW-physical interaction. Whereas, in the cases of Li_2S_x ($x=1,2,4,6,8$) as represented in figure 4(b-f), the interaction is mainly reduced as the number of sulfur-atoms increases, and a net charge transfer happens to h-BAs nanosheet from Li_2S_x polysulfides. Furthermore, for Li_2S_x ($x=1,2$) molecules adsorbed on the BAs system, the

depletion zone of electrons is in between molecules and h-BAs surface and the accumulation zones are on the h-BAs surface. This shows a net charge transfer happens from the molecules to the h-BAs monolayer. However, in the case of Li_2S_x ($x=4,6,8$), the accumulation zone of electrons are in between molecules (especially Lithium atom) and BAs system (B-atom), which stands for electron sharing instead of electron transfer.

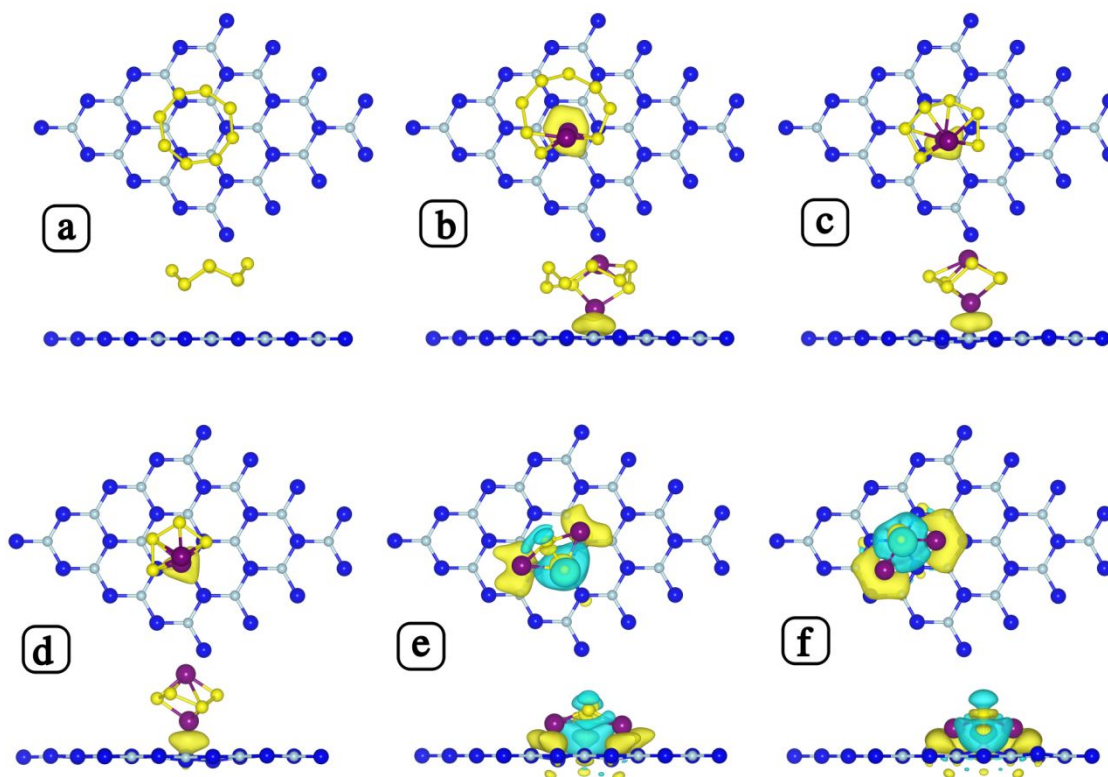


Figure. 4. Top and side views of deformed charge density between the S-containing species and h-BAs surface, at an iso-value of $1.5 \times 10^{-3} \text{ e/Bohr}^3$. (a) S_8 , (b) Li_2S_8 , (c) Li_2S_6 , (d) Li_2S_4 , (e) Li_2S_2 and (f) Li_2S . Yellow/blue indicate electron accumulation/depletion.

To further quantify these details, the Bader charge analysis is utilized, which states that the Li_2S_x molecules transferred charge to the h-BAs surface. In a non-lithiated S_8 molecule, a net charge transfer between different Sulfur atoms is about -0.02 |e| , and practically no-charge transfer can be obtained from molecule to BAs surface (0.00 |e|). This demonstrates the absence of any chemical interaction and confirms the findings discussed above. Moving to lithium sulfides, in the case of Li_2S_x ($x=8,6,4,2,1$) molecules, the charge transfer becomes higher with the reduction of S-atoms, causing a relatively stronger chemical bond. The net amount of charges 0.09 |e| , 0.21 |e| , 0.38 |e| , 0.52 |e| and 0.67 |e| are displaced from Li_2S_x ($x=8,6,4,2,1$) to h-BAs surface, respectively. In the case of Li_2S_x ($x=2,1$) adsorption, the Li- and S-atom make a principal contribution to adsorption energies with a transfer of charge about -0.04 |e| and 0.63

[e], respectively. This calculated charge transfer obtained from Bader's analysis is summarized in table 1.

3-3. Electronic properties of polysulfides adsorbed h-BAs surface

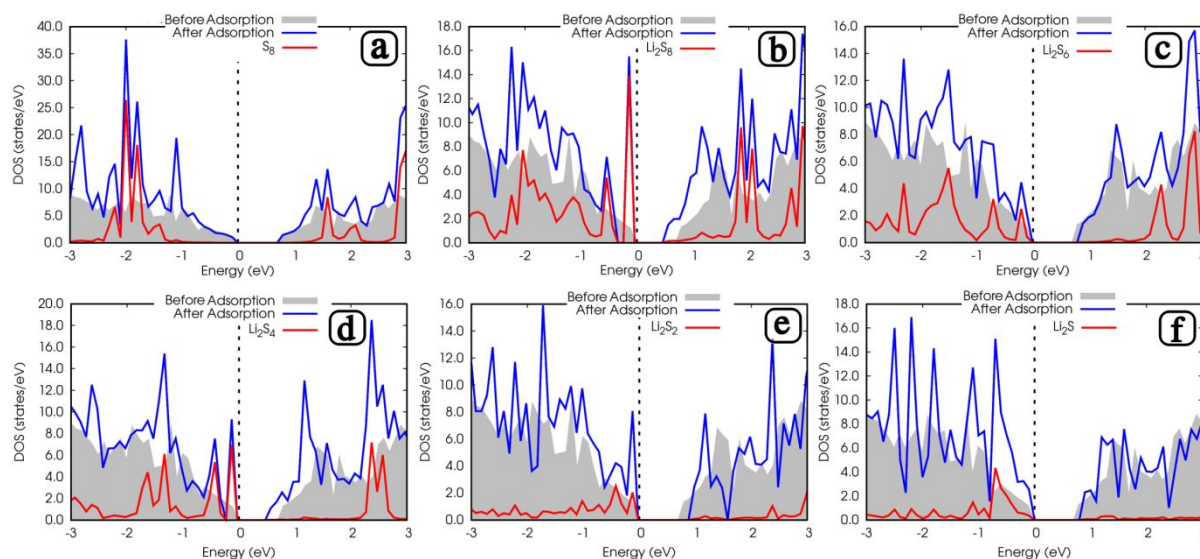


Figure. 5. Total Density of State (TDOS) of pristine h-BAs monolayer before and after the binding of (a) S_8 and (b-f) polysulfides Li_2S_x ($x=8,6,4,2,1$). The shaded area shows the DOS of the pristine BAS system, the blue color line stands for the total DOS of the molecule and surface. Red color depicts the projected DOS on the respective molecules during adsorption process.

Another crucial benchmark that significantly governs the efficiency of Li-S battery is the excellent electronic-conductivity of sulfur cathode host materials, that greatly affects the performance and operability of Li-S battery. As shown in the PDOS with bandstructure of the pristine h-BAs in figure S2, It is evident that the pristine h-BAs exhibits a semiconducting character with a direct bandgap of 0.76 eV in GGA-PBE calculation and 1.20 eV with HSE06 hybrid functional method. These results agree with previous studies^{34,35,49}. Additionally, based on the total and projected DOS plots. The low electrical-conductivity of S-containing molecules represents the central issue in Li-S batteries. To better explore the behavior of the h-BAs monolayer as a host material for Li-S batteries, it is indispensable to clearly understand the influence on the electronic properties after the adsorption of S_8/Li_2S_x molecules. Firstly, upon the adsorption of the non-lithiated S_8 molecule in figure 5a, no evident change is seen in the electronic structure owing to the low vdW interaction. There are no states around the Fermi level due to S_8 adsorption, this is in good coherence with the adsorption energy and charge transfer which is significantly low in comparison to other Li_2S_x molecules. In figure 5(b-f)

density of state for Li_2S_x molecule adsorption on BAs surface are presented with total DOS of pristine surface and projected states on the Li_2S_x molecules. It is pretty much clear that all these molecules have peak just below the fermi level which suggests the charge transfer from the molecule to the surface and also present stronger binding energies in comparison of non lithiated molecules S_8 . If we divide Li_2S_x molecules in two groups same as in previous section Li_2S_x ($x=4, 6, 8$) and Li_2S_x ($x=1, 2$), then we see the from the figure 2 and 4 that the later one adsorbed chemically with both the Lithium lying down on the surface, whereas first group adsorbed with only one lithium facing the surface. We can also conclude from the DOS picture in figure5 (e-f) that the strong interaction gives stronger hybridization of molecular states with VBM of the BAs surface, which reflects the stronger binding and higher charge transfer from the Li_2S and Li_2S_2 molecules to the BAs surface. On the other hand in figure 5(b-d), there are sharp molecular peaks in the pDOS which suggests less binding and further the lower charge transfer from molecule to the surface. Overall the molecular states below the Fermi suggests the charge transfer from all these molecules to the surface, and stronger hybridization of molecular states stands for more chemical nature of bonds in Li_2S and Li_2S_2 cases.

Addition to this, in the cases of Li_2S_x (with $x=8,6,4$), when adsorbed on h-BAs surface, the band gaps are reduced to 0.57 eV, 0.81 eV, and 0.54 eV, respectively (see figure 5(b-f)). This trend of reduction in the bandgap is favorable for the electro-conductivity of the host material on the basis of the formula:

$$\sigma = \exp(-\Delta E/K_bT) \quad (8)$$

where ΔE denotes the bandgap. Nevertheless, upon the adsorption of Li_2S_2 and Li_2S , there is a slight increase in the bandgaps of the system to 0.92 eV and 0.87 eV, respectively. A similar trend of increasing bandgaps is also observed in siligraphene (g-SiC_2) monolayer³¹, and it can readily yield the electrons directly to the redox-reaction of Li_2S_2 and Li_2S molecules.

3-4. Diffusion and catalytic decomposition of Li_2S on BAs surface

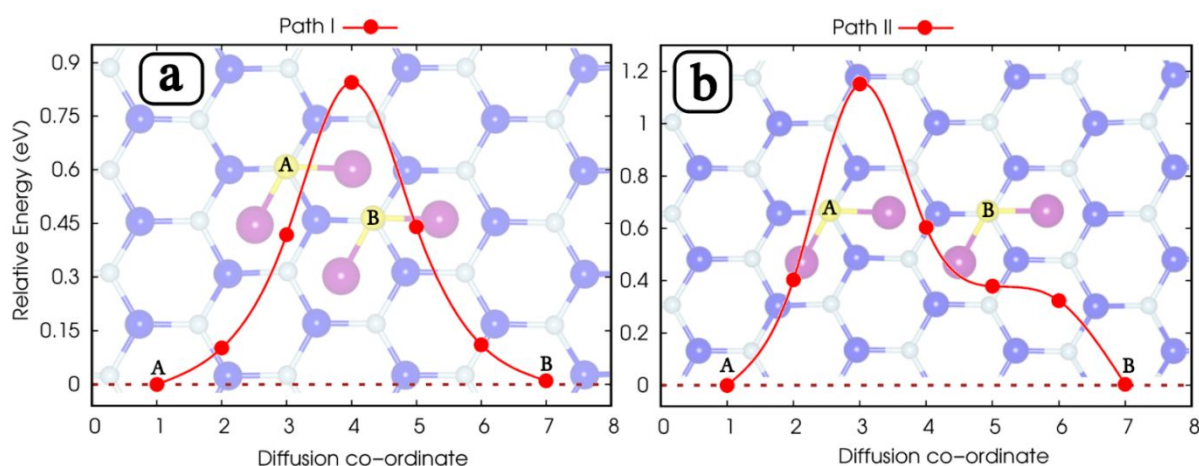


Figure. 6. Minimum-energy path (MEP) with their associated energy profiles for Li_2S over the surface of the h-BAs, (a) Pathway I, and (b) Pathway II. Starting at the most-stable binding site, and ending at a comparable binding site in the neighboring primitive cell.

The Li_2S diffusion barrier on h-BAs monolayer is symmetrically skipping the suitable and stable site (Both Li-atom situated at hollow site and S-atom above the B-atom) to another through two different pathways. We have used the CI-NEB approach to investigate the barrier height for the diffusion pathways. Both of these pathways and their corresponding migration energy barriers are shown in Figures 6a and b. In our calculation, the Li_2S diffusion barrier height is assessed to be 0.81 eV for pathway I and 1.15 eV for pathway II. Which suggests that the suitable pathway for Li_2S diffusion is the pathway I on h-BAs surface. Additionally, the diffusion barrier of the long-chain LiPSs, that constitute the highest-available long-chain LiPSs produced in the discharge process., i.e., Li_2S_4 , Li_2S_6 , and Li_2S_8 . The minimum energy path with corresponding energy profiles for the diffusion of long-chain polysulfides is illustrated in Figure. 7 (a–f). It's clear from these results that the barriers to diffusion in the pathway I are smaller, with energy profile of 0.14 eV, 0.26 eV, and 0.34 eV for Li_2S_x ($x=4,6,8$), respectively. It is evident from above that these values are nearly similar and quite small to that of BP monolayer (0.854 eV and 1.69 eV)²³. Based on a comparison of these values with previous reports on 2D materials, it can be stated that the 2D h-BAs nanosheet allows a fast diffusion of Li_2S and Li_2S_x ($x=4,6,8$) and avoids the shuttle effect.

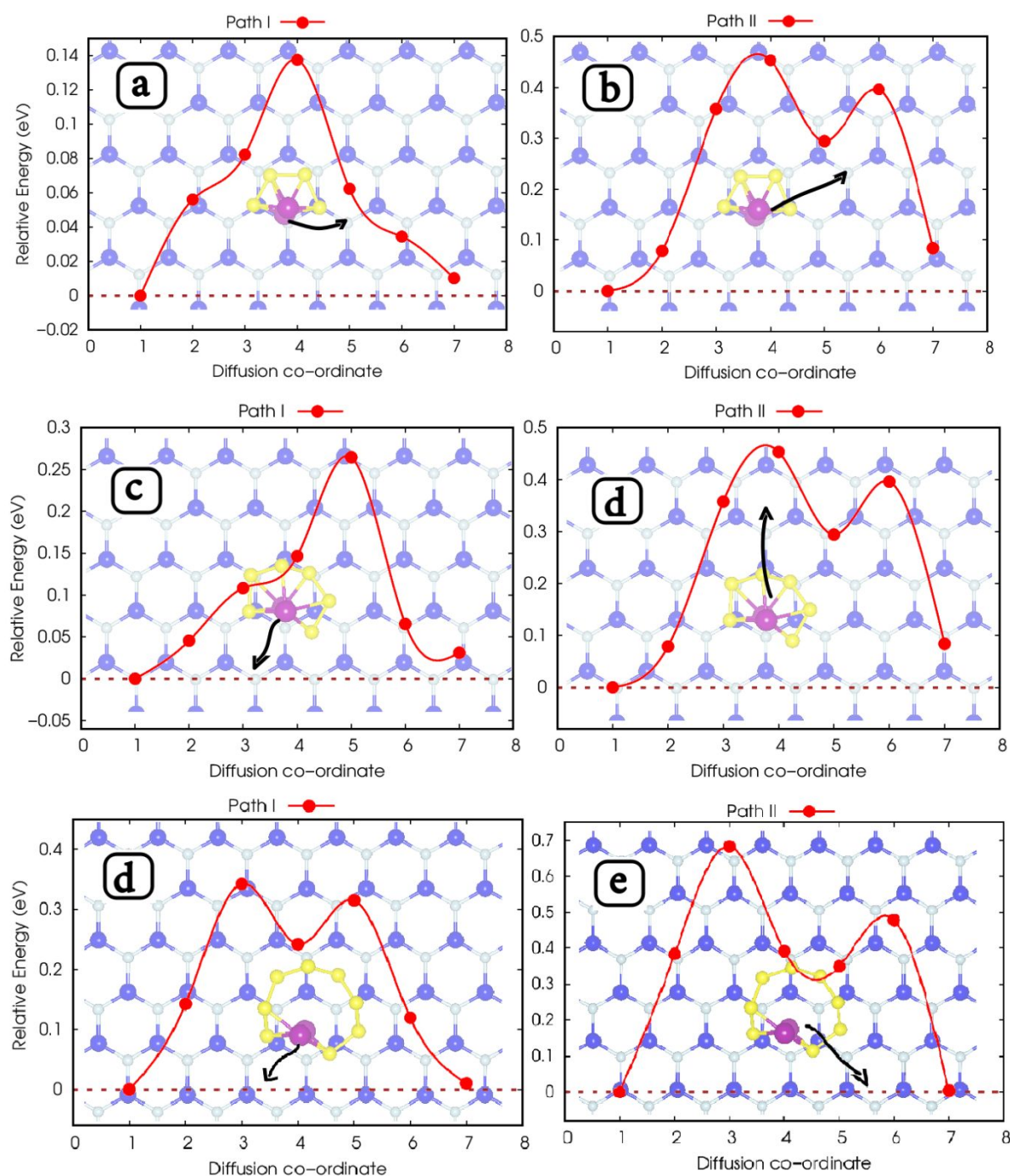
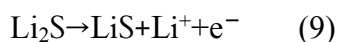


Figure 7. MEPs with corresponding energy profiles for (a, b) Li_2S_2 , (c, d) Li_2S_6 and (e, f) Li_2S_8 over the surface of the h-BAs monolayer in Pathway I and II, starting and ending at the most suitable adsorption site and a comparable site in the neighboring primitive cell.

In addition to the previously discussed diffusion barrier of Li_2S and large LiPSs, the Li_2S end-discharge polysulfide separates and gets transformed into the S_8 species after the separation of Li-S chemical-bond through the charge cycle. In spite of this, It is worthwhile to evaluate the effect of the h-BAs monolayer in the catalytic-oxidation of Li_2S polysulfide related with the

following reaction :



In other words, the extra-energy is necessary for cleaving the Li-S bond in order to release the Li-atom, which further diffuses towards its neighboring favorable binding site. Figure 8 illustrates the minimum-energy path with the corresponding relative energy of the decomposition processes on the h-BAs system. The decomposition of a Li_2S molecule is defined according to the bonding energy of Li-atoms on h-BAs surface. The decomposition energy barrier estimated is about 1.31 eV. However, if we compare the Li_2S catalytic decomposition on h-BAs to some other promising materials highlighted in previous studies, it makes our argument more compelling. It is obvious from the above that this value is smaller in comparison to those of several promising two-dimensional materials such as graphene⁵⁵ and TiCS_2 ⁵⁶ with decomposition barrier of 1.81 eV and 1.51 eV, respectively.

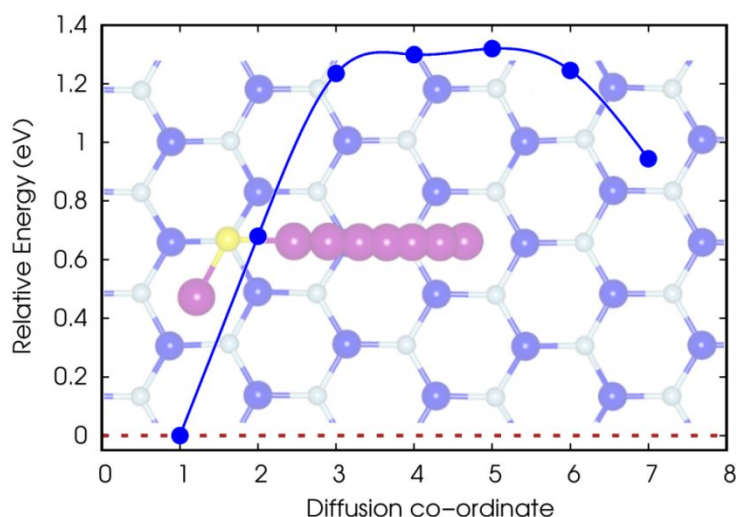


Figure. 8. Minimum-energy path (MEP) and corresponding energy profile for the decomposition of a Li-S bond during intercalation on a h-BAs monolayer.

3-5. Adsorption of Li_2S_4 , Li_2S_6 and Li_2S_8 species on the 2D vacancy BAs

Taking into account the many 2D graphene-like materials for Li-S based batteries, it is worth addressing the following question: Is it possible to improve the anchoring effects of 2D h-BAs monolayer for Li-S batteries by introducing vacancies, dopant, and mechanical strain? For this purpose, the interaction of three different S-containing molecules (Li_2S_8 , Li_2S_6 , and Li_2S_4) and

defective (vacancies) h-BAs monolayers were comprehensively investigated. Three different vacancies are taken into account in this study, as presented in Figure 9a, Boron-vacancy (V_B), Arsenic-vacancy (V_{As}), and a Boron-Arsenic divacancy (V_{B-As}). It is found, after full relaxation of the three-defected systems, that all B- and As-atoms remain at the same Z-coordinates and the bond-lengths d_{B-As} around the vacancies have a negligible change in V_B and V_{As} monovacancies and but during the di-vacancy V_{B-As} , bond length get increased with ~ 0.5 Å as illustrated in Figure S7 in SI. To assess the stability of these defect systems, the calculated formation energies based on the equation (3) was 7.99 eV, 6.28 eV, and 9.91 eV for V_{As} , V_B , and DV, respectively. Our results for these three vacancies show that the formation energy of V_B is smaller than that of V_{As} and DV, suggesting that V_B is more energetically favorable and should be frequently observed on the pristine h-BAs system. The thermal stability of the three vacancies is checked with MD-simulations at 300K temperatures. The Figure S8 in SI illustrate the change in the total energy of these vacancies systems and the snapshots, indicating that there was no B-As bond breaking and no distortion of the structures, which can also be seen in table S2 and the movies C1-3 in the SI.

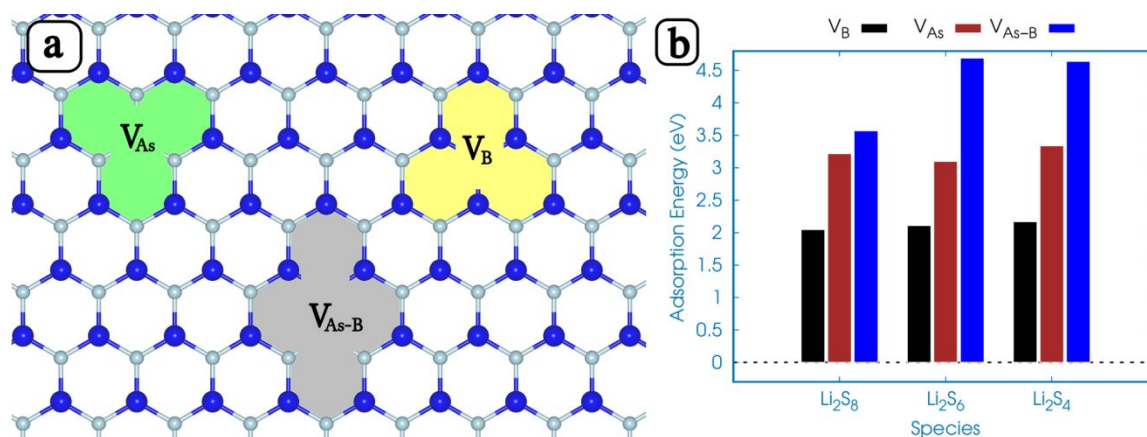


Figure. 9. (a) A schematic representation of different vacancies in h-BAs monolayer and (b) corresponding calculated binding energies of Li_2S_8 , Li_2S_6 , and Li_2S_4 species on the defected system of h-BAs monolayers.

Furthermore, it was found that by introducing vacancies, the adsorption energies of Li_2S_8 , Li_2S_6 , and Li_2S_4 LiPSs adsorbed on h-BAs monolayer had been improved significantly. As revealed in Figure. 9b and Table. 2, the V_B exhibits only a minor improvement in the adsorption energies of Li_2S_8 , Li_2S_6 , and Li_2S_4 (2.05 eV, 2.11 eV and 2.17 eV) compared to pristine h-BAs monolayer (1.07 eV, 0.83 eV and 0.77 eV), DOL (0.78 eV, 0.81 eV and 0.79 eV) and DME (0.95 eV, 1.16 eV and 0.96 eV). In the case of V_{As} and di-vacancy (DV), the E_{ads} of chosen

LiPSs molecules resulted in higher binding energy 3.00 eV, suggesting an optimum chemical-interaction for Li-S batteries. In the case of all three defected systems, as illustrated in Figure S9, both Li-atoms of the chosen Li_2S_8 , Li_2S_6 , and Li_2S_4 were parallel to the h-BAs surface, and the entire geometries of LiPSs are slightly changed. Thus, for instance, in the case of Li_2S_6 , the one Li-atom is chemically bonded to As-atoms in V_B and DV systems, while the strong chemical bond S-B are formed in the case of V_{As} system. Here, we can also see that the defective h-BAs sheet represents the enhancement in binding strength of Li_2S_4 , Li_2S_6 , and Li_2S_8 as compared to LiPSs with electrolytes. The creation of vacancies is one of the most efficient methods for improving the binding strength of LiPSs clusters for Li-S batteries, which is an important governing factor for cathode materials. Consequently, the defective system of the h-BAs surface has been proved to be considerably good anchoring materials for Li-S batteries.

Table. 2. Comparison of calculated adsorption energies (E_{ads}) of LiPSs on the pristine and defective system of h-BAs monolayers, binding energies (E_b) of LiPSs with electrolyte molecules (DOL and DME liquid organic electrolytes), respectively.

	E_{ads} (Pristine)	E_{ads} (V_B)	E_{ads} (V_{As})	E_{ads} (V_{B-As})	E_b (DOL)	E_b (DME)
Li_2S_8	1.07	2.05	3.57	3.22	0.78	0.95
Li_2S_6	0.83	2.11	4.69	3.10	0.81	1.16
Li_2S_4	0.77	2.17	4.64	3.34	0.79	0.96

4. Conclusion

The prospective of Li-S batteries for outperforming today’s LIBs with substantial progress has been a real challenge that lies in finding suitable electrode materials. In hindsight, we have observed a rapid development in the exploration of structure optimizations and electronic characteristics of 2D h-BAs monolayer as a source of energy storage applications per se in battery technologies. The feasibility of using a 2D h-BAs system as a host material for Li-S batteries has been explored. Summarizing our findings, the illustrated properties of pristine h-BAs nanosheets proved to be a suitable electrode material owing to its more considerable binding energy of all different S-containing molecules on the h-BAs surface, which is between 0.47–3.55 eV. The anchoring mechanisms have been addressed by examining the van der

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waals physisorption and chemisorption contribution. Additionally, the adsorption of S_8/Li_2S_x LiPSs reduces the bandgap of the h-BAs monolayer in some cases, displaying improved efficiency of the electronic conductivity. The ultra-fast energy barriers and catalytic decomposition of Li_2S over the surface imply high diffusion of Li_2S on the h-BAs surface. Our results showed excellent promising performance with improved electronic conductivity suggesting h-BAs monolayer as a suitable electrode material for Lithium sulfur batteries, which potentially offers guidance to experimentalists about exploring and developing the capabilities of this 2D material in energy storage systems.

Acknowledgments

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Supplementary Information

The Supplementary Information available: Calculated electronic density of states, Band structure using PBE-GGA and HSE06 functional, Dynamical stability of the system using phonopy, the thermal stability of system before and after LiPSs adsorption using AIMD at 300K, calculated Li-S and S-S bonds lengths of S_8 and Li_2S_x ($x=1,2,4,6,8$) before and after adsorption, optimized molecules of LiPSs, S_8 and organic liquid electrolytes DOL and DME molecules, The physical and chemical interactions of LiPSs and S_8 with DOL and DME, optimized defective systems, the thermal stability of defective systems at 300K and the Molecular dynamic evolution of the systems.

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